

S385 - TMA03

Investigating how Baryon Acoustic Oscillations depend on Redshift in the Dark Energy
Survey Galaxy Catalogue

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1 Abstract

This report aims to investigate the presence and evolution of Baryon Acoustic Oscillation (BAO) signals across different redshift bins using galaxy data from the Dark Energy Survey (DES) catalogue. By analysing the two-point angular correlation function, $\omega(\theta)$, I examined the presence of a BAO 'bump' imprinted across multiple redshift bins. The measured angular size, θ_{BAO} , was then compared with theoretical predictions to. The results confirmed that the BAO scale follows the expected trend, although limited by a weak or absent BAO signal, which was improved at specific redshift ranges with optimised rebinning. Despite this, the bins considered were consistent with theoretical data and support the use of BAO as a cosmological probe.

2 Introduction

Baryon Acoustic Oscillations (BAOs) are a relic of sound waves that propagated through the Photon-Baryon plasma in the early Universe, leaving an imprint in the large-scale structure of galaxy distribution. As described in Chapter 7 [1], these oscillations, arising from competition between radiation pressure and gravitational attraction, froze into the matter distribution when the universe transitioned from Radiation to Matter-Dominated at the epoch of the last scattering, ($z \approx 1090$). The resulting characteristic scale, the sound horizon, is directly related to the BAO scale, and represents the maximum distance sound waves travelled in the Photon-Baryon plasma before recombination. This provides a standard ruler for measuring cosmic distances, making BAOs one of the key observational probes for studying universe expansion.

The BAO signal is observed as a slight excess in galaxy clustering at a preferred comoving separation, which appears as a 'bump' in the two-point angular correlation function $\omega(\theta)$ as described in chapters 10 [2] and 11 [3]. This statistical measure provides a likelihood for finding galaxy pairs at a given angular separation, allowing us to gather information on the growth and evolution of the Universe's expansion rate. Measuring the BAO angular size at different redshifts helps us determine the evolution of dark energy and the geometry of the universe, critical for validating the standard Cold Dark Matter model [4].

3 Methods

To begin the investigation, I started by choosing an initial set of redshift slices to analyse. Given that our data spans redshifts $0.60 < z < 1.10$, I chose the following bins to ensure a thorough and broad range of redshifts was covered:

- $0.60 < z < 0.70$
- $0.70 < z < 0.80$
- $0.75 < z < 0.85$
- $0.80 < z < 0.90$
- $0.85 < z < 0.95$
- $0.90 < z < 1.00$
- $0.95 < z < 1.05$
- $1.00 < z < 1.10$

The binning was refined through iteration, settling on the same bin width as in Practical Cosmology 3, giving a good balance of signal strength and statistical significance. The selection made gave a sufficient galaxy count in each bin to obtain meaningful measurements of the two-point correlation $\omega(\theta)$, and spanned across the entire dataset.

To detect the BAO signal, I calculated $\omega(\theta)$ to quantify the clustering of galaxies as a function of angular separation, θ . This allowed me to then analyse the function and look for the presence of a distinct 'bump' in the distribution to indicate the presence of BAOs. The calculation of $\omega(\theta)$ was based on this formula from Chapter 11 [3]

$$\omega(\theta) = \frac{\langle P(\theta) \rangle}{R(\theta)} - 1 \quad (1)$$

where $P(\theta)$ is the probability of observing two galaxies separated in the sky by an angle between θ and $\theta + \delta\theta$ and $R(\theta)$ is the probability of observing the galaxies with the same angular separation if they were uniformly distributed throughout the universe. However, to make the results more concise with real data, I used the Landy-Szalay Estimator formula, expressed as:

$$\omega(\theta) = \frac{DD(\theta) - 2fDR(\theta) + f^2RR(\theta)}{f^2RR(\theta)} \quad (2)$$

Where:

1. $DD(\theta)$ represents the number of pairs of galaxies in the catalogue with angular separation θ and $\theta + \delta\theta$.
2. $RR(\theta)$ contains 'fake' galaxies that are uniformly distributed throughout space and represents the number of pairs of galaxies in the synthetic catalogue with angular separation θ and $\theta + \delta\theta$.
3. $DR(\theta)$ represents the number of pairs of galaxies with angular separation θ and $\theta + \delta\theta$ if one is taken from the real catalogue and one from the synthetic.
4. f is the number of galaxies in the observed catalogue divided by the number of galaxies in the synthetic catalogue.

Once the values of $\omega(\theta)$ were computed for each redshift bin, I was able to plot them on a logarithmic scale against an array representing the centres of bins in θ for which the correlation function was calculated. The plots all looked roughly linear, characteristic with a power law function. To make the BAO signatures more pronounced, I rebinned the data, which helped eliminate any noise and fluctuations in each bin, making the BAO feature more visible. For $0.80 < z < 0.90$, this appeared as such:

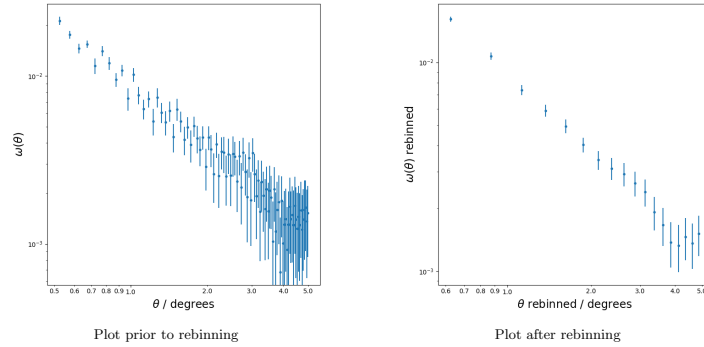


Figure 1: Improving clarity of BAO signal

We can see that the data follows a falling power law trend for most of the θ range, with a small deviation at the 'bump', and some flattening as larger θ values are approached. Based on this, the data was fit using two components, a broken power law component, and a Gaussian component (to model the shape of the BAO signal).

Our combined model was therefore:

$$f_{combined}(\theta, \Theta) = f_{BPL}(\theta) + f_{BAO}(\theta) \quad (3)$$

The model fitting process involved selecting parameter estimates based on visual inspections of my plots and then refining them through trial and error. The χ^2 minimisation provided a good evaluation of the fit and was used to create an initial model for the calculated $\omega(\theta)$ points, that aligned reasonably well with the data. I plotted the rebinned correlation function, initial parameter model, and broken power law continuum to check the fitting.

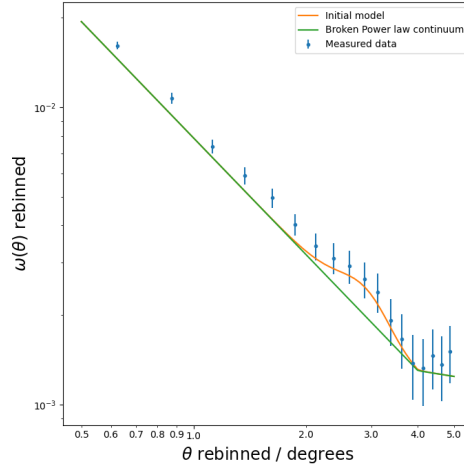


Figure 2: Initial Parameters

The next step in the process was setting up parameter bounds through normalisation in order to obtain a best fitting model. The normalisation constants were adjusted through iteration, and the rebinned correlation function was then plot with the model overlaid to obtain a best fit plot on a set of axes.

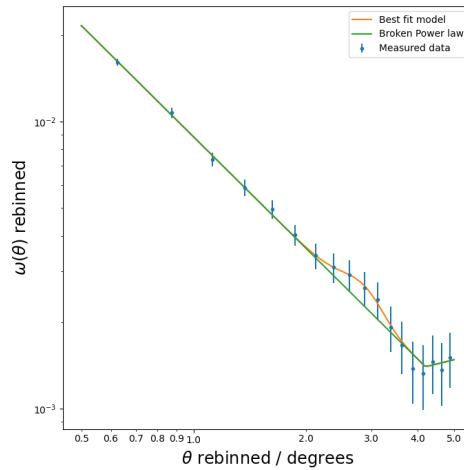


Figure 3: Best Fit model

Once a model passing through all data points or their error bars was obtained, I proceeded to check whether the overlaid Gaussian peak aligned well with the BAO features. This was done through removal of the continuum component from the data and overlaying the plot with the Gaussian component to obtain the following:

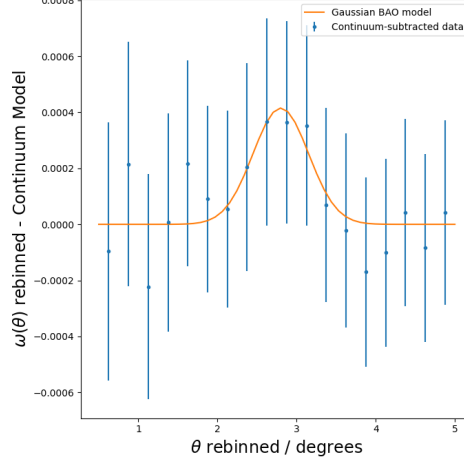


Figure 4: Gaussian model overlay

Given that there were only 5/6 bins forming the peak of the model, we see just how weak the BAO signal really is. Despite this, it is clear the model fits the data well, and that the extracted value of θ_{BAO} from the correlation function is reliable. We now need to check the uncertainty associated with our value. I did this by investigating how the fit quality changed as θ_{BAO} diverged from its best fit value. I plotted the curve of χ^2 versus θ_{BAO} , checking whether the lowest point on the curve was at the global best fit BAO angular size, and if the curve showed a roughly parabolic shape across the range of uncertainty values for θ_{BAO} .

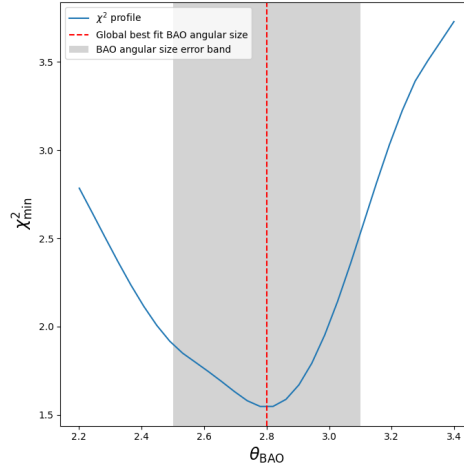


Figure 5: Uncertainty estimate

As shown in Figure 5, the best fit BAO angular size and its uncertainty appear reasonable and reliable, with the shape of the χ^2 profile aligning with all above mentioned criteria. Confirming all our values and uncertainties concluded the investigation, and allowed me to compare the fitted angular size of the BAO signal to theoretical predictions for how it should evolve with redshift. This comparison gave insight into the consistency of obtained measurements, and the validity of using Baryon Acoustic Oscillations as a standard ruler in cosmology.

To compare the results with theory, I calculated how the physical size of the structure evolved with redshift. It was assumed that growth is completely dominated by the Hubble expansion, and therefore the physical size could be described entirely in terms of the comoving size and the scale factor:

$$d_{p,BAO}(z) = a(z)d_{c,BAO} \quad (4)$$

Using the fact that $a = (1 + z)^{-1}$:

$$d_{p,BAO}(z) = \frac{d_{c,BAO}}{1 + z} \quad (5)$$

Once the expected physical size at a given redshift of the structures leading to the BAO signal was obtained, I was able to use the angular diameter distance corresponding to the redshifts to calculate the expected angular size for each redshift bin. This calculation was only valid assuming the angular size of the BAO signal was small enough to use the small angle approximations.

$$\theta_{BAO}(z) = \frac{d_{p,BAO}(z)}{d_A(z)} \quad (6)$$

To present all findings, for each redshift bin, I plotted the theoretical prediction against the measurements made. I did this by calculating a mean redshift value for the galaxies in each bin selected and plotting the theoretical angular sizes on the same axes as our measured values and their associated error bars to check for agreement.

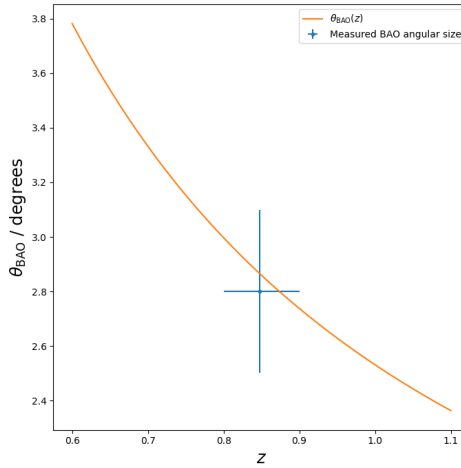


Figure 6: Theoretical vs Measured vales

In this case we can see that the data obtained aligns with theoretical vales, as the error bars overlap with the model line.

I repeated this for all other chosen bins, and for those displaying a visible BAO signal, I plotted the final BAO angular sizes against the theoretical predictions on one plot. This was done by combining the variable list for each bin into a dictionary, and saving it into a csv file, allowing me to create a final plot by recalling the data from each file and plotting the points obtained from the saved lists.

4 Results

The results of this investigation provide an analysis of the BAO signal across different redshift bins using the two-point correlation function $\omega(\theta)$. The function was initially assessed visually, to determine whether a BAO induced 'bump' was observable. Following this, model fitting was performed to obtain values for θ_{BAO} .

As discussed above, I chose a range of redshift bins spanning the entire dataset being analysed. I did however encounter some difficulty at the lower end of the set. After computing $\omega(\theta)$ and rebinning data, the plots still appeared entirely linear, and any attempts to fit the data were unsuccessful, ultimately

concluding that the BAO signal was either too weak to be detected, or did not exist for such low redshifts. This led me to exclude the $0.60 < z < 0.70$ and $0.70 < z < 0.80$ redshift bins from the final plot.

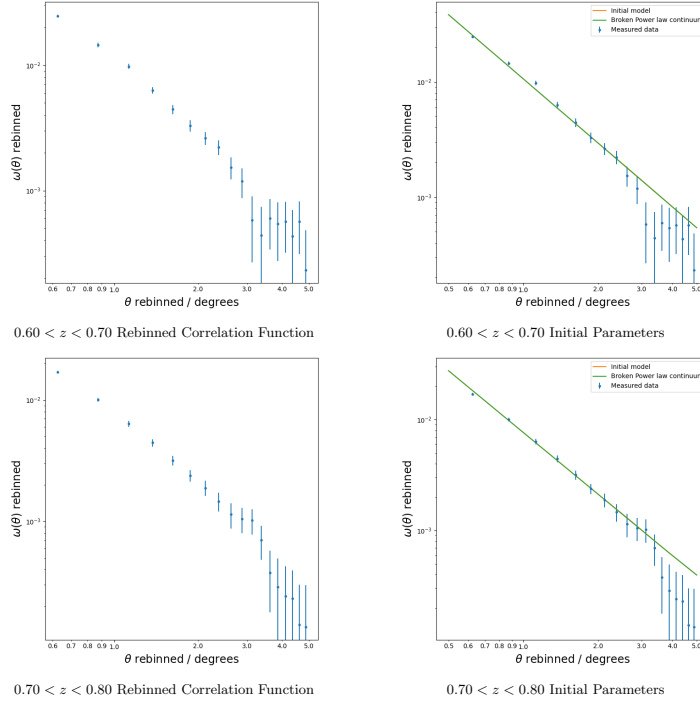


Figure 7: Excluded Redshift bins

For the remainder of my chosen redshift bins, I was able to successfully locate a BAO 'bump' in my rebbinned correlation plots. The plots all demonstrated a promising BAO signature, and although the strength of the signal varies across bins, all were visible and I felt confident moving onto the model fitting.

The six rebbinned correlation functions appeared as follows:

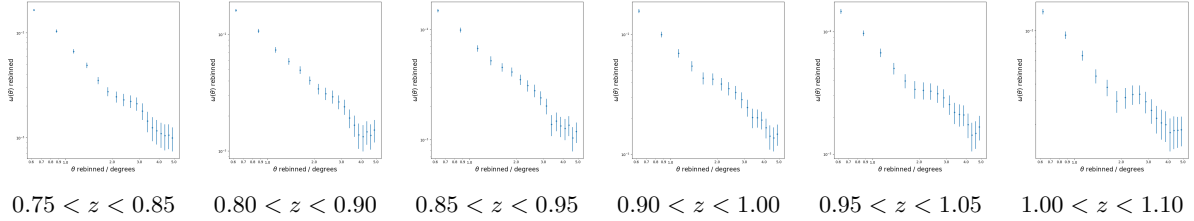


Figure 8: Rebbinned Data Plots

The fitting process involved applying parameters to the broken power law and best fit model, followed by the normalisation and application of bounds to further refine the fit. This ensured that when plotted, the model captured the trend by passing through each data point or remaining within its corresponding error bar.

My models of best fit are:

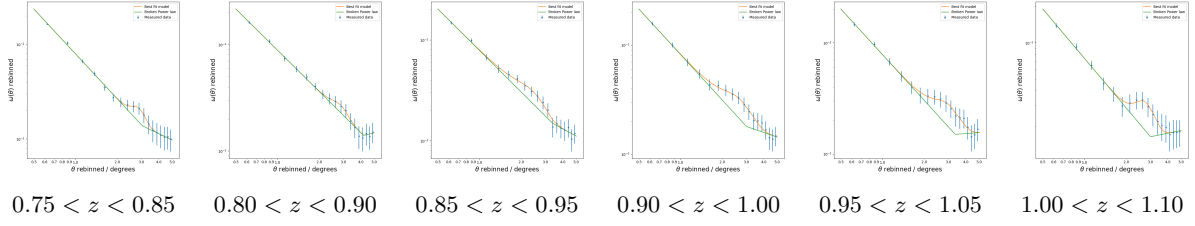


Figure 9: Models of best fit

Since all plots of best fit looked promising, I overlaid them onto the Gaussian model to assess the BAO feature alignment, and compare this to the best fit BAO angular sizes against the χ^2 profile for each redshift bin to evaluate the reliability of the uncertainty estimates.

The resulting plots were:

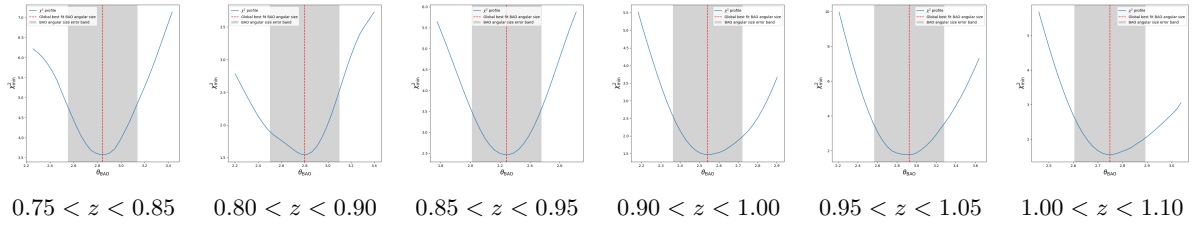


Figure 10: Uncertainty estimates compared to χ^2 profile

Figure 10 clearly demonstrates that the uncertainty values for all redshift bins are within acceptable limits, as evidenced by the parabolic nature seen throughout the plots. The smooth curvature of the χ^2 profile indicated well constrained fits, confirming the validity of the measured BAO angular size. Using this information, I was able to plot the relationship between θ_{BAO} and z for each bin, as follows:

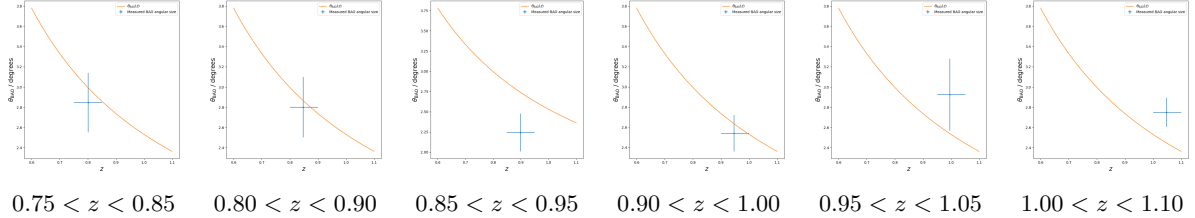


Figure 11: Theoretical vs Measured value individual plots

While some of the values, specifically those in the $0.85 < z < 0.95$ and $1.00 < z < 1.10$ bins do not intersect the theoretical overlay curve, the overall trend aligns with predictions. To illustrate the relationship between the extracted BAO angular sizes and theoretical expectations, I combined all the individual results into one final plot, illustrated below:

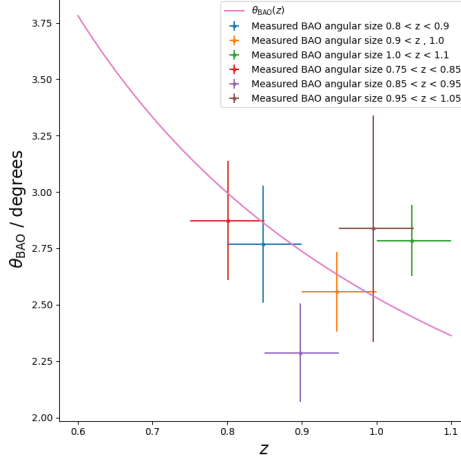


Figure 12: Final Plot

Figure 12 illustrates the measured BAO angular sizes across the valid redshift bins, ranging $0.75 < z < 1.10$, overlaid with the theoretical BAO scale evolution. There is a clear decreasing trend as redshift increases, with most data points closely conforming to the theoretical curve. There are minor deviations which may be attributed to statistical uncertainties in measuring $\omega(\theta)$ or systematic error within the dataset. Despite this, the majority of data points align well with theoretical predictions within their uncertainties, except the $0.85 < z < 0.95$ and $1.00 < z < 1.10$ points, for which neither of their error bars cross over the curve. The outliers and large uncertainties for some redshift bins may result from lower noise-to-signal ratio in $\omega(\theta)$, which can make it more difficult to precisely locate the BAO peak and find θ_{BAO} .

5 Discussion and conclusions

This investigation aimed to analyse the BAO signal across different redshift bins, using galaxy data from the DES. By computing $\omega(\theta)$, I was able to assess if a BAO signal was detectable, and see how its angular size evolves with redshift. The results gathered confirm that the BAO angular size is comparable with theoretical data and follows the expected trend, decreasing with evolving redshift. The final plot successfully illustrates that the detected values of θ_{BAO} were within the expected uncertainties, reinforcing the use of BAOs as a standard ruler and supporting the cold dark matter model. While the mid range bins provided a very clear BAO signal, there were several limitations, particularly with lower redshifts exhibiting an undetectable BAO signal. For future work, I would increase sample size to reduce noise, and refine the bins further to optimise width and parameters.

Overall, I am very happy with the data obtained and believe there to be enough evidence to conclude the detected θ_{BAO} align with expectations. I think that obtaining more data for the lower redshift bins to allow us to see a BAO signal would however be very beneficial, as we do not have any points plotted prior to $z \approx 0.80$, putting into question the validity of the model at low redshift values.

References

- [1] Chapter 7. *Cosmology with the Cosmic Microwave Background*. Cosmology Part 2, Pages 29-47.
- [2] Chapter 10. *Structure Formation and the Cosmic Web*. Cosmology Part 2, Pages 109-114.
- [3] Chapter 11. *Formation of Stars and Galaxies*. Cosmology Part 2, Pages 129-157.
- [4] Chapter 9. *Early Universe physics meets observations*. Cosmology Part 2, Page 107.